

A Closure-Stable Banach-Target Converse to the Bornological Operator Identity

Candidate partial answer to Remark 5.6 of arXiv:math/0506183

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Status. Candidate substantial partial result, likely valid, subject to human review. The converse is proved for Banach targets whose induced ideal bornology is stable under norm closure. The unrestricted locally convex and non-closure-stable cases remain open.

1 Source question

Remark 5.6 (PDF p. 18) of Wong, *The triangle of operators, topologies, bornologies*, [1], fixes an operator ideal $\mathcal{A}$ on locally convex spaces and its generated bornology $\mathcal{M} = \mathcal{B}(\mathcal{A})$. It observes that $\mathcal{A}$ -bornologicality of Y implies

$$\mathcal{L}^b(X, Y) \cap \mathcal{O}(\mathcal{M})(X, Y) = \mathcal{O}^b(\mathcal{M})(X, Y) \quad \text{for every locally convex space } X, \quad (1)$$

and asks whether the converse holds.

Here $T \in \mathcal{L}^b(X, Y)$ means that some zero-neighborhood has topologically bounded image; $T \in \mathcal{O}(\mathcal{M})(X, Y)$ means that T sends every bounded subset of X to an $\mathcal{M}(Y)$ -bounded set; and $T \in \mathcal{O}^b(\mathcal{M})(X, Y)$ means that some zero-neighborhood has $\mathcal{M}(Y)$ -bounded image.

2 Result

Call $\mathcal{M}(Y)$ *norm-closure stable* if the norm closure of every $\mathcal{M}(Y)$ -bounded set is again $\mathcal{M}(Y)$ -bounded.

Proof intuition

Test the operator identity on the underlying vector space of Y equipped with its strongest locally convex topology. Its bounded sets live in finite-dimensional subspaces, so the identity map automatically sends all bounded sets to ideal-bounded sets. On the other hand, the Banach unit ball is an absorbing disk and hence a neighborhood for the strongest topology; the identity is therefore locally bounded. The assumed equality forces some absorbing disk to be ideal-bounded. Closure stability makes its norm closure ideal-bounded, and Baire category upgrades that closed absorbing disk to a norm neighborhood. It then absorbs every norm-bounded set, exactly giving $\mathcal{A}$ -bornologicality.

Theorem 1 (Banach, closure-stable converse). *Let $\mathcal{A}$ be an operator ideal on locally convex spaces, $\mathcal{M} = \mathcal{B}(\mathcal{A})$, and let Y be a Banach space. Assume:*

1. $\mathcal{M}(Y)$ is norm-closure stable;

Proof. By Theorem 5.1 if X is $\mathfrak{A}$ -topological then $\mathcal{O}^b(\mathcal{P})(X, Y) = \mathcal{L}^b(X, Y)$ and $\mathcal{O}(\mathcal{P})(X, Y) = \mathcal{L}(X, Y)$. The equality follows. Conversely, assume the equality holds for every LCS Y . It suffices to show that $\mathcal{L}(X, N) \subseteq \mathcal{O}^b(\mathcal{P})(X, N)$ for each normed space. Let $N_{\mathcal{P}}$ be the LCS given by equipping N with the $\mathcal{P}(N)$ -topology. By the functorial property (GT_2) of $\mathcal{P}$, any T in $\mathcal{L}(X, N)$ also belongs to $\mathcal{L}(X_{\mathcal{P}}, N_{\mathcal{P}}) = \mathcal{O}(\mathcal{P})(X, N_{\mathcal{P}})$. By the hypothesis, $T \in \mathcal{O}^b(\mathcal{P})(X, N_{\mathcal{P}})$. Since $\mathcal{P}(N)$ is compatible with the dual pair (N, N') by (GT_1) , we have $T \in \mathcal{O}^b(\mathcal{P})(X, N)$. It follows the desired assertion. $\square$

Remark 5.6. If we let $\mathcal{M} = \mathcal{B}(\mathfrak{A})$ then a LCS Y being $\mathfrak{A}$ -bornological implies

$$\mathcal{L}^b(X, Y) \cap \mathcal{O}(\mathcal{M})(X, Y) = \mathcal{O}^b(\mathcal{M})(X, Y)$$

for each LCS X . We do not know if the converse is true.

Let $\mathfrak{A}$ be an operator ideal on LCS's. Denote by $\mathfrak{A}_{\mathfrak{B}}$ the operator ideal defined on Banach

Figure 1: The preceding topological proof and the bornological converse question in Remark 5.6 of the source (PDF p. 18).

2. identity (1) holds for every locally convex space X .

Then Y is $\mathcal{A}$ -bornological.

Proof. Let $Y_{\max}$ denote the vector space underlying Y , equipped with the strongest locally convex topology, and let

$$J : Y_{\max} \longrightarrow Y$$

be the identity map. Every seminorm on the vector space is continuous for the strongest topology, so J is continuous. The closed norm unit ball B_Y is absolutely convex and absorbing, hence it is a zero-neighborhood in $Y_{\max}$. Its image is norm bounded. Therefore $J \in \mathcal{L}^b(Y_{\max}, Y)$.

We next show that $J \in \mathcal{O}(\mathcal{M})(Y_{\max}, Y)$. A bounded subset of a vector space with the strongest locally convex topology is contained in a finite-dimensional subspace and is bounded there. Indeed, if a bounded set contained an infinite linearly independent sequence (x_n) , extend it to a Hamel basis and assign weights tending to infinity to the x_n ; the resulting seminorm is continuous for the strongest topology and is unbounded on the set. The finite-dimensional assertion then follows from the usual equivalence of finite-dimensional norms.

Every finite-dimensional bounded subset of Y is $\mathcal{M}(Y)$ -bounded: realize it inside a scalar multiple of the image of a Euclidean unit ball under a finite-rank map, and recall that every operator ideal contains all finite-rank operators. Thus J sends bounded subsets of $Y_{\max}$ to $\mathcal{M}(Y)$ -bounded subsets.

Apply (1) with $X = Y_{\max}$. We obtain $J \in \mathcal{O}^b(\mathcal{M})(Y_{\max}, Y)$. Consequently there is an absolutely convex zero-neighborhood U in $Y_{\max}$ such that $U = J(U)$ is $\mathcal{M}(Y)$ -bounded. Every zero-neighborhood is absorbing, so

$$Y = \bigcup_{m=1}^{\infty} mU \subseteq \bigcup_{m=1}^{\infty} m\overline{U} = Y,$$

where closure is in the Banach norm. By norm-closure stability, $\overline{U}$ is $\mathcal{M}(Y)$ -bounded. The sets $m\overline{U}$ are norm closed; Baire's theorem implies that one of them has nonempty interior. Since $\overline{U}$ is absolutely convex, it is therefore a norm zero-neighborhood.

It follows that $B_Y \subseteq r\overline{U}$ for some $r > 0$. Hence B_Y is $\mathcal{M}(Y)$ -bounded. Every norm-bounded subset of Y is contained in a scalar multiple of B_Y , so the ordinary Banach bornology and the $\mathcal{A}$ -bornology coincide. This is precisely that Y is $\mathcal{A}$ -bornological. $\square$

3 Examples and scope

The closure hypothesis is satisfied for several standard ideal bornologies on Banach targets:

- finite-dimensional bounded sets (the finite-rank ideal);
- relatively norm-compact or precompact sets;
- relatively weakly compact sets.

Thus, in each of these cases, the identity in Remark 5.6 characterizes the corresponding $\mathcal{A}$ -bornological Banach spaces.

The proof isolates the obstruction to the unrestricted question. Without closure stability, an $\mathcal{M}$ -bounded absorbing disk need not absorb the Banach unit ball uniformly; it may behave like the unit ball of an everywhere-defined discontinuous stronger norm. The argument above does not show that its norm closure remains in $\mathcal{M}(Y)$. Nor does it cover a general nonnormable target, because such a space has no topologically bounded absorbing disk with which to make the identity from the strongest topology locally bounded.

Upgrade attempts and novelty search. Eight focused attempts checked formal dualization, strongest-topology and mixed-topology constructions, Baire upgrades, common ideals, and possible non-closed counterexamples. The attempt record explains why the closure-free step remains unresolved. Bounded searches on 17 August 2026 covered the run indexes, exact title and arXiv id, Remark 5.6, the verbatim open sentence, and arXiv queries for the displayed operator identity. No later explicit answer was found.

Human-review recommendation. Verify the characterization of bounded sets in $Y_{\max}$, the use of finite-rank maps to place them in $\mathcal{M}(Y)$, and norm-closure stability for any specific ideal to which the theorem is applied. The main classification question is whether closure stability can be removed or violated by a genuine ideal bornology.

References

- [1] N.-C. Wong, *The triangle of operators, topologies, bornologies*, arXiv:math/0506183v1 (2005), Remark 5.6.